

Target classification with low-resolution radar based on dispersion situations of eigenvalue spectra

CHEN Feng*, LIU HongWei, DU Lan & BAO Zheng

National Laboratory of Radar Signal Processing, Xidian University, Xi'an 710071, China

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Abstract Most low-resolution radar systems, especially ground surveillance radar systems, work at relatively low pulse repeat frequency (PRF) and with short time-on-target (TOT) (duration in scanning). Low PRF leads to Doppler ambiguity and short TOT results in low Doppler resolution, which poses a problem to target classification with low-resolution radar based on the jet engine modulation (JEM) characteristic of radar echo. From the pattern classification viewpoint, we propose a method of using dispersion situations of JEM eigenvalue spectra to categorize aeroplanes into three kinds, namely turbojet aircraft, prop aircraft and helicopter. We analyze the mathematical model of JEM echoes consisting of a series of line spectra and regard them as a sum of several series of harmonious waves. Classification features can be extracted based on the harmonious wave sum model. Some schemes for extracting features from echoes within or between pulses are proposed. Low-dimensional features are extracted to reduce computation burden. Our methods do not compensate for the fuselage echoes and are insensitive to the variation of fuselage Doppler. The feasibility of our methods is demonstrated by simulation experiment.

Keywords jet engine modulation (JEM), low-resolution radar, low pulse repeat frequency (PRF), short time-on-target (TOT), target classification

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1 Introduction

Most conventional air defense radar systems belong to low-resolution radar systems. To make conventional radar provide target category information besides detecting them will be substantially significant for our territory security. In theory, this can be achieved by combination of low-resolution radar that detects targets and high-resolution radar that classifies them; however, if the two tasks could be accomplished just by low-resolution radar in use, the radar system complexity can be efficiently reduced through saving broad band radar equipment. Practically, the low-resolution radar at present has very limited classification capability. Because workers have to decide target category and distinguish multiple planes flying in a raid cluster by naked eyed characteristic of received echoes' intensity and fluctuation of intensity, target moving trajectories, target velocity, target acceleration, friend or foe response signals, etc. Note that this totally depends on the proficiency of workers, which does not satisfy the requirements of modern battle. Hence, finding practical classification methods for low-resolution radar is an important

*Corresponding author (email: chenfeng6120@163.com)

research area with widespread promising applications. Nevertheless, on account of conventional radar characteristics of low PRF, narrowband and short TOT, in fact, for a long time, target classification or identification has been a tough task.

Hitherto, feature extraction methods in literature available for low-resolution radar target classification can be summarized as two categories: one uses received echoes' RCS [1], intensity and fluctuation of intensity [2], fluctuation of echoes' phases [2], motion parameters (target altitude [3], flying velocity, flying acceleration), time-spectrum (the dynamic trends that target space position and motion state vary with time is referred to as time-spectrum), and visual effect and two-dimensional gray image based features, to decide target category, while the other important one is based on the periodical modulated echoes produced by the high-speed rotating structures (rotating wings, fans or props) in the aeroplane's impetus parts, i.e., JEM characteristics [4–9]. These structures have modulated Doppler spectra located as line spectra on both sides of the continuous carrier frequency, and the modulated wave's periodicity, the modulated Doppler line spectra's interval are determined by the blade number and the rotation speeds of the rotating elements. The above properties are invariant to the radar line of sight angle, i.e., they agree with any radar azimuth as long as there is no shielding of the rotating structures. Most current methods of extracting JEM features are first compensating for the fuselage component signal, and then estimating the modulated wave's periodicity and the modulated Doppler line spectra's interval by parametric methods, such as complex cepstrum method [10], auto correlation method [11], periodogram method [12], AR model power spectrum method [13]. Ref. [11] used neural net to estimate JEM spectrum. However, this kind of parametric spectrum estimation methods hardly achieves high precision because conventional radar cannot supply sufficiently high PRF or sufficiently long observation time.

This paper focuses on the feature extraction method of JEM characteristics with low-resolution radar. In all efforts to sidestep the contradiction between the conventional radar's real condition that is low PRF and short TOT, and the requirement of parametric spectrum estimation methods, high RPF and long TOT, we propose some schemes to extract features by sample points within or between pulses from the point of pattern classification view. Our methods accomplish the aim of classifying the three kinds of aeroplanes (turbojet aircraft, prop aircraft and helicopter) without compensating for the fuselage signal. This paper first indicates that the number of the modulated line spectra is the proper characteristic that is insensitive to low PRF and short TOT. In order to explain this point, we introduce the mathematical model of the rotating structures with low-resolution radar and the ideal modulated characteristics, including modulated periodicity, the modulated line spectra's interval and number. Then, starting from the harmonious wave sum model, we propose our classification methods, deduce the relationship between the eigenvalue spectrum and the modulated line spectrum number under the condition of short pulses series, discuss the effects on the eigenvalue spectrum of low PRF and short TOT, and put forward some schemes to extract features using sample points within or between pulses by eigenvalue decomposing (EIG), and for the latter, a four-dimensional feature is extracted to reduce computation burden. Finally, we analyze the effect of the fuselage component on eigenvalue spectrum, and conclude that our features can be used without compensating for the fuselage component.

2 Mathematical model of rotating structures

Without considering of the ground clutter and other electronic jamming, the complex envelope received by low-resolution radar's quadrature receiver consists of at least the fuselage component, the rotating structure components (JEM components), and noise component. We have (see [14, 15])

$$s(t) = a_s(t)[C_f S_f(t) + C_{\text{JEM}} S_{\text{JEM}}(t)] + C_n w_n(t), \quad (1)$$

where $a_s(t)$ is the overall limitation of air-defense radar system to JEM characteristics, equivalent to $a_s(t) = a_t(t)a_r(t)a_\alpha(t)$, $a_t(t)$ being the transmitter signal model, $a_r(t)$ the receiver model, $a_\alpha(t)$ the antennas scanning model. $S_f(t)$, $S_{\text{JEM}}(t)$ and $w_n(t)$ are the fuselage, the rotating structure and the noise returned echoes, respectively. C_f , C_{JEM} and C_n are the intensities of the fuselage, the rotating structure and the noise component returned echoes, respectively. Supposing there are M ideal rotating structures

at the far field distance R_0 to radar, azimuth angle α and pitching angle β , the rotating structure echoes of the fundamental frequency are (see [5, 7, 14, 15])

$$\begin{aligned}
 S_{\text{JEM}}(t) = & \exp\left(-j2\pi f_d t - j\frac{4\pi R_0}{\lambda}\right) \sum_{m=0}^{M-1} V(\alpha, \beta', \ddot{P}) \\
 & \cdot \sum_{k=0}^{N_m-1} [g_{1_m} + g_{2_m}(\theta_{k_m} + \omega_{r_m} t - \alpha)](L_{2_m} - L_{1_m}) \\
 & \cdot \text{sinc}\left[\frac{2\pi}{\lambda}(L_{2_m} - L_{1_m}) \cos \beta' \cos(\theta_{k_m} + \omega_{r_m} t - \alpha)\right] \\
 & \cdot \exp\left\{-j\frac{4\pi}{\lambda}\left[\frac{L_{2_m} + L_{1_m}}{2} \cos \beta' \cos(\theta_{k_m} + \omega_{r_m} t - \alpha)\right]\right\}. \quad (2)
 \end{aligned}$$

In eq. (2) the m th rotating structure's rotation speed is ω_{r_m} , the distances from the bottom and the top of a blade to the center of blades are L_{1_m} , L_{2_m} respectively, M is the number of rotating wings, fans or props, N_m is the m th rotating structure's blade number, the fuselage Doppler frequency is $f_d = 2v/\lambda$, where v is the projection of fuselage velocity onto the radar line-of-sight. $\theta_{k_m} = \theta_0 + 2\pi k/N_m$, $k = 0, 1, 2, \dots, N_m - 1$, where θ_0 is the 0th blade's initial rotating angle. Omitting blades' twist, ϑ_m denotes blade angle, $g_{1_m} = \sin(|\beta'| + \vartheta_m) + \sin(|\beta'| - \vartheta_m)$, $g_{2_m} = \text{sign}(\beta') \cdot [\sin(|\beta'| + \vartheta_m) - \sin(|\beta'| - \vartheta_m)]$. When the rotating plane of the rotating structures is parallel to the aeroplane's flying direction, $\beta' = \beta$; otherwise, when they are vertical, $\beta' = \pi/2 - \beta$. Visible function $V(\alpha, \beta', \ddot{P})$ is used to express the shielding problem of the rotating structures, where $\ddot{P}$ denotes a certain plane.

Figure 1 illustrates the geometrical relationship between radar and a rotating fan when rotating plane parallels the flying direction. When the aeroplane flies from far away to the point above the radar along the azimuth angle α , the rotating plane is vertical to the radar line-of-sight. This is analogous to the case where the rotating plane is vertical to the flying direction. Hence, vertical rotating structures at distance R_0 , azimuth angle α and pitching angle β are equivalent to parallel rotating structures at distance R_0 azimuth angle α , and pitching angle $\pi/2 - \beta$.

We define the ideal JEM characteristics as JEM characteristics under sufficiently high PRF and sufficiently long TOT conditions. Those characteristics mainly include the periodicity at time domain, the line spectrum interval, the spectrum bandwidth and line spectrum number at Doppler domain. Eq. (2)'s Fourier Transformation is as follows [5, 7, 14, 15]:

$$F_{\text{JEM}}(f) = \sum_{m=0}^{M-1} V(\alpha, \beta', \ddot{P}) \sum_{n=-N_{1_m}}^{N_{1_m}} c_{m,n} \delta(f - f_d + n f_{T_m}). \quad (3)$$

Thus, the modulated spectra consist of several series of line spectra. For the m th rotating structure, the line spectrum interval $f_{T_m} = q N_m f_{r_m}$ depends on the blade number N_m and the blade velocity $f_{r_m} (\omega_{r_m} = 2\pi f_{r_m})$. N_{1_m} is the line spectrum number of a single side. For even blades, $q = 1$; for odd blades, $q = 2$. The amplitude of line spectrum $C_{m,n}$ is determined by parameters $\lambda, L_{2_m}, L_{1_m}, \beta', N_m, \theta_0$ and Bessel function. The line spectrum number N_{1_m} and bandwidth B_{1_m} of a single side are figured out as follows [5, 7, 14, 15]:

$$N_{1_m} = \frac{8\pi L_{2_m} \cos \beta'}{N_m \lambda}, \quad (4)$$

$$B_{1_m} = \frac{8\pi q f_{r_m} L_{2_m} \cos \beta'}{\lambda} = \frac{4\omega_{r_m} q L_{2_m} \cos \beta'}{\lambda}. \quad (5)$$

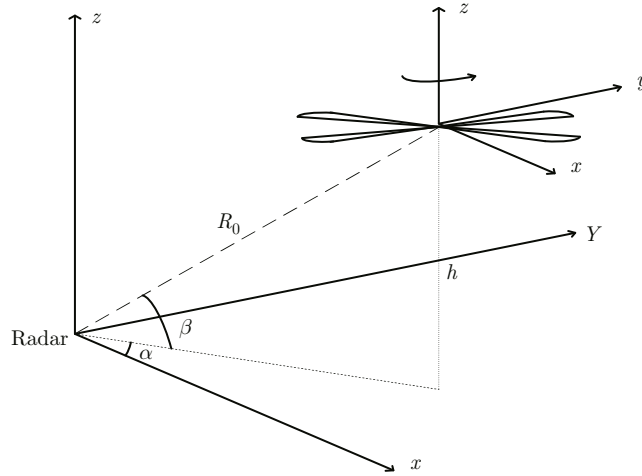


Figure 1 Geometrical relationship between radar and rotating structure when rotating plane parallels the flying direction.

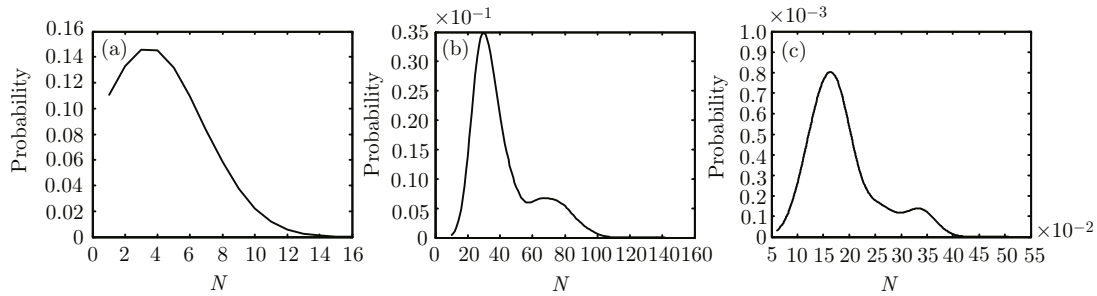


Figure 2 The estimated probability densities of the two-side line spectrum number of the three kinds of 25 earoplanes in typical scene. (a) Turbojets; (b) prop aircrafts; (c) helicopters.

In the related literatures, JEM feature extraction methods concentrated on the aspect of estimating line spectrum interval [10–19]. In fact, low-resolution radar usually illuminates a target for a very short time (often 20–100 ms) during a single sweep; as a consequence, the resolution of Doppler frequency is very low. Especially, when TOT is less than the rotating structure's periodicity, we just estimate an unstable instantaneous spectrum. At the same time, low resolution radar's PRF usually ranges from several hundred to several thousand Hertz, which means that the three kinds of aeroplanes' JEM echoes are in general ambiguous in Doppler domain. The ambiguity numbers are related to the aeroplane parameters f_{rm} , L_{1m} , L_{2m} , radar parameter λ , the pitching angle β , etc. Therefore, for radar characterized as short TOT and low PRF, it is arduous to realize accurate line spectrum interval estimation, even accurate spectrum bandwidth estimation. We notice that the line spectrum numbers of the three kinds of aeroplanes are very discriminant under typical scene. Figure 2 shows the estimated probability densities of the number of the two-side line spectra of rotating structures of the three kinds of 25 aeroplanes in typical scene (which will be described in section 4), where the two-side line spectrum number is $2N_{1m} - 1$, and N_{1m} can be obtained from eq. (4). Figure 2 indicates that the orders of harmonious wave number in the three kinds of aeroplanes' rotating structures are quite different. Accordingly, in the following section, we will extract JEM line spectrum number based on a harmonious wave sum mathematical model.

3 Feature extraction of JEM characteristics with low-resolution radar

According to eq. (3), omitting the fuselage components echoes, the JEM modulated echoes are the sum of several series of equal interval harmonious waves. Let the total number of harmonious waves be K .

Then the returned echoes of radar is

$$x(n) = S_{\text{JEM}}(n) + w(n) = \sum_{i=1}^K \alpha_i \exp[j(\omega_i n + \phi_i)] + w(n), \quad n = 0, 1, \dots, N-1, \quad (6)$$

where $w(n)$ is the complex Gaussian noise with zero mean and covariance σ^2 . We suppose that the signal is weak stationary, and the initial phases ϕ_i , $i = 1, 2, \dots, K$, follow a uniform distribution on the interval $[0 \ 2\pi]$. Since the noise and the JEM signal are uncorrelated, the returned echoes' covariance matrix is

$$\mathbf{R}_{xx} = \mathbf{R}_{ss} + \sigma^2 \mathbf{I} = \frac{1}{N-p} \sum_{i=1}^K W_i e_i e_i^H + \sigma^2 \mathbf{I}, \quad (7)$$

where W_i is the power of the i th harmonious wave, p is the length or order of the sample, $e_i = \{1 \exp(\omega_i) \exp(2\omega_i) \cdots \exp[(p-1)\omega_i]\}$ is the signal vector. Let $p > K$. Then $p \times p$ matrix $\mathbf{R}_{ss}$ is singular. Thereby, if we figure out the eigenvalues and eigenvectors of $\mathbf{R}_{ss}$, and describe them by $\lambda_m, \nu_m, m = 1, 2, \dots, p$, where eigenvalues are in decreasing order, then the last $p-K$ eigenvalues are zeros. This can lead to the following equation:

$$\mathbf{R}_{xx} = \sum_{m=1}^K (\lambda_m + \sigma^2) \nu_m \nu_m^H + \sum_{m=K+1}^p \sigma^2 \nu_m \nu_m^H. \quad (8)$$

Accordingly, $\nu_m, m = 1, 2, \dots, p$ are also the eigenvectors of $\mathbf{R}_{xx}$. There are $p-K$ smaller eigenvalues in $\mathbf{R}_{xx}$'s eigenvalues, corresponding to eigenvectors constituting the noise space, and K bigger eigenvalues, corresponding to eigenvectors constituting the signal space. Obviously, the harmonious wave number can be obtained by estimating K , the number of bigger eigenvalues in the p eigenvalues. Nevertheless, the previous condition $p > K$ can hardly be guaranteed by low-resolution radar at low PRF and with a short TOT. Consequently, we discuss the case $p < K$ in the following.

We firstly introduce the following theorem [20] and its proof is in Appendix A.

Let $\mathbf{T}_n = \{\kappa_{i,j} | \kappa_{i,j} = \xi_{j-i}, \xi_{-j} = \xi_j^*\}$ be an Hermite Toeplitz matrix, where $i = 1, 2, \dots, n, j = 1, 2, \dots, n$. Let $\mathbf{T}_k (k = 1, 2, \dots, n-1)$ be its k th order principle submatrix, and $P_k(\lambda) = \det(\mathbf{T}_k - \lambda \mathbf{I}_k)$, $q_1(\lambda) = \xi_0 - \lambda$, $q_k(\lambda) = P_k(\lambda)/P_{k-1}(\lambda)$, $c_k = (\xi_{-1}, \xi_{-2}, \dots, \xi_{-k})^T, k = 1, 2, \dots, n$.

Theorem. If λ is not one of $\mathbf{T}_{n-1}$'s eigenvalues, and $y_{n-1}(\lambda)$ is the solution vector of eq. (9),

$$(\mathbf{T}_{n-1} - \lambda \mathbf{I}_{n-1}) y_{n-1}(\lambda) = c_{n-1}, \quad (9)$$

then

$$q_n(\lambda) = \xi_0 - \lambda - c_{n-1}^H y_{n-1}(\lambda), \quad (10)$$

$$q'_n(\lambda) = -1 - y_{n-1}^H y_{n-1}(\lambda). \quad (11)$$

And if λ is one of the eigenvalues of $\mathbf{T}_n$, its corresponding eigenvector is $x_n(\lambda) = \begin{bmatrix} -1 \\ y_{n-1}(\lambda) \end{bmatrix}$.

According to the above theorem, if we let λ be one of the eigenvalues of $\mathbf{T}_n$ and not one of the eigenvalues of $\mathbf{T}_{n-1}$, and $y_{n-1}(\lambda)$ be the solution vector of equation $(\mathbf{T}_{n-1} - \lambda \mathbf{I}_{n-1}) y_{n-1}(\lambda) = c_{n-1}$, $\mathbf{T}_{n-1}$'s eigenvectors be $\lambda_1, \lambda_2, \dots, \lambda_{n-1}$, and their corresponding eigenvectors be $Q_1, Q_2, \dots, Q_{n-1}$, and $\mathbf{Q} = [Q_1, Q_2, \dots, Q_{n-1}]$. Then

$$\begin{aligned} \mathbf{T}_{n-1} - \lambda \mathbf{I}_{n-1} &= \mathbf{Q}^H \Sigma \mathbf{Q}, \\ \Sigma &= \text{diag}(\lambda_1 - \lambda, \lambda_2 - \lambda, \dots, \lambda_{n-1} - \lambda). \end{aligned} \quad (12)$$

Because $y_{n-1}(\lambda)$ is the solution vector of $(\mathbf{T}_{n-1} - \lambda \mathbf{I}_{n-1}) y_{n-1}(\lambda) = c_{n-1}$ and λ is not one of the eigenvalues of $\mathbf{T}_{n-1}$, we can get $\det(\mathbf{T}_{n-1} - \lambda \mathbf{I}_{n-1}) \neq 0$. Thus

$$y_{n-1}(\lambda) = (\mathbf{T}_{n-1} - \lambda \mathbf{I}_{n-1})^{-1} c_{n-1},$$

$$q_n(\lambda) = \xi_0 - \lambda - c_{n-1}^H y_{n-1}(\lambda) = \xi_0 - \lambda - c_{n-1}^H (\mathbf{T}_{n-1} - \lambda \mathbf{I}_{n-1})^{-1} c_{n-1}. \quad (13)$$

Obviously, $P_n(\lambda) = \det(\mathbf{T}_n - \lambda \mathbf{I}_n) = 0$, $q_n(\lambda) = P_n(\lambda)/P_{n-1}(\lambda) = 0$. Hence

$$\lambda = \xi_0 - c_{n-1}^H \mathbf{Q} \Sigma^{-1} \mathbf{Q}^H c_{n-1} = \xi_0 - \frac{|\xi_{-1}|^2}{\lambda_1 - \lambda} - \frac{|\xi_{-2}|^2}{\lambda_2 - \lambda} - \dots - \frac{|\xi_{-n+1}|^2}{\lambda_{n-1} - \lambda}. \quad (14)$$

The n roots of the above equation with respect to λ are eigenvalues of $\mathbf{T}_n$. Multiply both sides of eq. (14) by $(\lambda - \lambda_1)(\lambda - \lambda_2) \cdots (\lambda - \lambda_{n-1})$, and put the right side's terms to the left side. Then λ is the root of eq. (15)

$$\begin{aligned} f(\lambda) = & \lambda(\lambda - \lambda_1)(\lambda - \lambda_2) \cdots (\lambda - \lambda_{n-1}) - \xi_0(\lambda - \lambda_1)(\lambda - \lambda_2) \cdots (\lambda - \lambda_{n-1}) \\ & - |\xi_{-1}|^2(\lambda - \lambda_2)(\lambda - \lambda_3) \cdots (\lambda - \lambda_{n-1}) - |\xi_{-2}|^2(\lambda - \lambda_1)(\lambda - \lambda_3) \cdots (\lambda - \lambda_{n-1}) - \dots \\ & - |\xi_{-n+1}|^2(\lambda - \lambda_1)(\lambda - \lambda_2) \cdots (\lambda - \lambda_{n-2}). \end{aligned} \quad (15)$$

Since $f(+\infty) > 0$, $f(\lambda_1) < 0$, $f(\lambda_2) > 0, \dots, f(-\infty) < 0$, the n roots fall in the ranges $(\lambda_1, +\infty)$, (λ_2, λ_1) , $\dots, (-\infty, \lambda_{n-1})$ respectively, in a decreasing order.

If we define the submatrix of $\mathbf{R}_{xx}$ obtained by deleting the same last q rows and columns as the $(p-q)$ th order principle submatrix of $\mathbf{R}_{xx}$, then full rank Hermite covariance matrix $\mathbf{R}_{xx}$'s eigenvalues can be "interpolated" into the eigenvalues of the $(p-1)$ th order principal submatrix of $\mathbf{R}_{xx}$, and eigenvalues of its $(p-1)$ th order principal submatrix can be "interpolated" into the $(p-2)$ th order principal submatrix's eigenvalues, and the eigenvalues of the remainder principal submatrices can be "interpolated" in a similar way.

When radar scans a target in a short time which results in limited sample points, the obtained eigenvalue spectrum (a vector containing eigenvalues in a decreasing order is identified by eigenvalue spectrum) is equivalent to "sampling" from the eigenvalue spectrum under sufficient observation time condition. The two eigenvalue spectra's shapes are alike, meaning that the new spectrum is discriminant potential in classifying the three kinds of aeroplanes. Let it be noted that the word "sampling" here is not exactly the same as that in sampling technology for signal processing. Actually, the ideal spectrum just provides several ranges for its principal matrices' eigenvalues; in other words, the "sampling" here is picking eigenvalues from several nonoverlapping ranges on the real number axis. Furthermore, when the order of one principal matrix is much larger than that of another, it is not quite sure whose corresponding eigenvalues are bigger, except the biggest and smallest ones. Figure 3 shows the eigenvalue spectra of a prop aircraft of the 37th and 35th orders. The former corresponds to observation time of 15 ms and the latter to 14 ms. As aforesaid, the biggest (or smallest) eigenvalues of the 37th order principal submatrix are bigger (or smaller) than those of the 35th's, which is in accordance with what is shown in Figure 3.

As we know, the low PRF means low sampling rate. Let $s(n)$ be the sample points of JEM continuous echoes $S_{\text{JEM}}(t)$, $t \in [0, T]$, when f_s is the sampling rate. Thus

$$x(n) = s(n) + w(n) = S_{\text{JEM}}(nf_s) + w(nf_s), n = 0, 1, 2, \dots, \lfloor T/f_s \rfloor, \quad (16)$$

where $\lfloor T/f_s \rfloor$ means to find the variable's nearest integer towards zero. If $f_s \geq f_{s^*}$, where f_{s^*} is the Nyquist sampling rate, for the case where a single observation time on target is long enough, the eigenvalue spectrum Λ of covariance matrix $\mathbf{R}_{xx}$ is ideal, from which the short TOT case is "sampling"; if $f_s < f_{s^*}$, down sampling will lead to Doppler frequency aliasing; however, the number of line spectra is almost unchanged unless a line spectrum coincidentally superposes on another, thus decreasing the number. But it rarely happens. Figure 4(a) shows the sample points of a continuous wave at the Nyquist sampling rate, and its Fourier transformation is in Figure 4(b). Figure 4(c) is the down sampling points of the same continuous wave, and Figure 4(d) is its Fourier transformation. In the middle of Figure 4(a) and (c) the sampling time points of them are sketched out, where Figure 4(a)'s sampling time points are above the time axis and Figure 4(c)'s are under the time axis. Comparing Figure 4(b) and 4(d), the down sampling Fourier Transformation is just the Nyquist sampling one folded at frequencies shown by the dash lines, during which their spectra width and line spectrum interval are reduced, but the number of line spectra

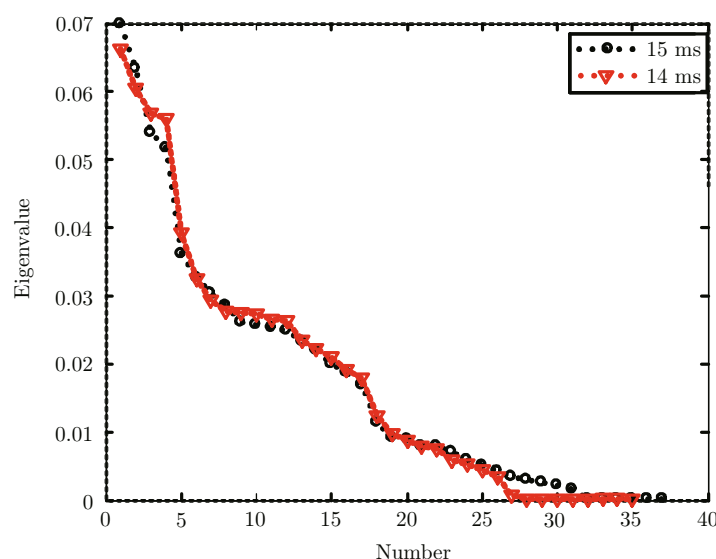


Figure 3 The eigenvalue spectra of the 37th order (15 ms), (14 ms) and 35th order principal submatrices of a prop aircraft.

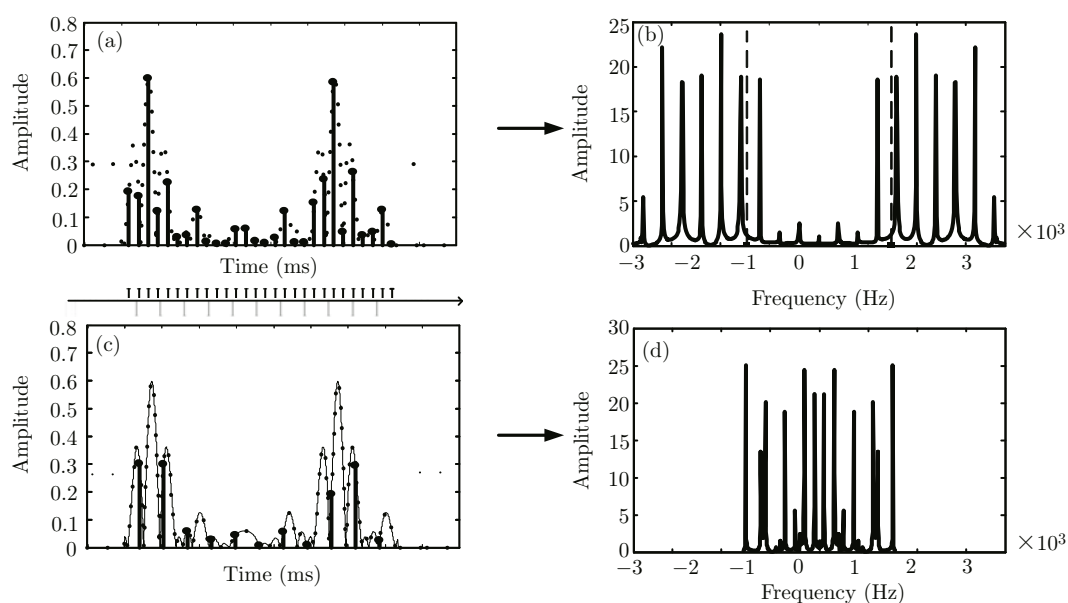


Figure 4 Sample points and their frequency spectra at the Nyquist sampling rate and down sampling rate, respectively.

are identical. Consequently, when radar works at down sampling rate, the corresponding eigenvalue spectrum Λ' contains the information of the line spectrum number, and when the observation time are shortened, the eigenvalue spectrum is a “sample” of Λ' . Thus, when radar works at low PRF with short TOT, the extracted eigenvalue spectra are of the same shape as the ideal eigenvalue spectra.

For a fixed order $p < K$, under the typical scene, helicopters' eigenvalue spectra are flattest since they contain the largest number of harmonious waves, prop aircrafts' eigenvalue spectra are flatter, and turbojets' eigenvalue spectra are sharpest. Figure 5 shows some examples of the three kinds of 25 aeroplanes' JEM eigenvalue spectra (where we randomly set the initial azimuth angle of aeroplanes to radar) while radar works at C band and PRF of 5 kHz, for observation time of 100 and 25 ms respectively. Since this radar can receive 500 pulses within 100 ms, the largest estimated harmonious wave number is 250, corresponding to the abscissa of Figure 5(a), if the covariance matrices are estimated by the samples obtained by adding a slipping window of half pulse length. Similarly, the eigenvalue spectra for 25 ms

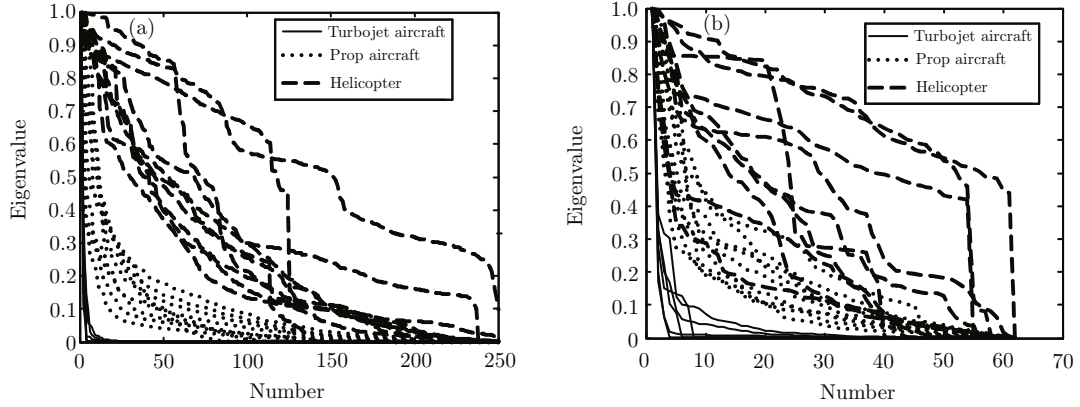


Figure 5 Typical JEM eigenvalue spectrum examples of the three kinds of 25 aeroplanes. (a) 100 ms; (b) 25 ms.

are of 62 dimension, corresponding to Figure 5(b)'s abscissa. For the purpose of clarity, eigenvalues are prearranged in a decreasing order and divided by their largest eigenvalues. Figure 5(a) indicates that the three kinds of aeroplanes' eigenvalue spectra are evidently discriminant, and Figure 5(b) indicates the eigenvalue spectra begin to overlap when the observation time is curtailed, though they still have discriminant potential. Known from Figure 2(a), (b) and (c), turbojets' and prop aircrafts' eigenvalue spectra contain all the harmonious waves for observation time of either 100 or 25 ms, but helicopters' eigenvalue spectra have a dimension that are far less than their harmonious wave number. Thus, in Figure 5 the eigenvalue spectra of helicopters in truth are the "sample" of the ideal eigenvalue spectra.

If low-resolution radar's sampling rates along range and Doppler are f_r and f_{dp} respectively, observation time is T , and pulse width is τ , then the number of sampling points along range and Doppler are $N_r = \lceil \tau \cdot f_r \rceil$, $N_{dp} = \lfloor T \cdot f_{dp} \rfloor$ respectively. We write the radar returned echo X as

$$X = \begin{bmatrix} x(0) & \cdots & x(n_{dp}/f_{dp}) & \cdots & x((N_{dp}-1)/f_{dp}) \\ \vdots & & \vdots & & \vdots \\ x(n_r/f_r) & \cdots & x(n_{dp}/f_{dp} + n_r/f_r) & \cdots & x((N_{dp}-1)/f_{dp} + n_r/f_r) \\ \vdots & & \vdots & & \vdots \\ x((N_r-1)/f_r) & \cdots & x(n_{dp}/f_{dp} + (N_r-1)/f_r) & \cdots & x((N_{dp}-1)/f_{dp} + (N_r-1)/f_r) \end{bmatrix}, \quad (17)$$

where $x(t) = \sum_{i=1}^K \hat{\alpha}_i \exp[j(\hat{\omega}_i t + \hat{\phi}_i)]$, $t \in [0, T]$, $n_r = 0, 1, \dots, N_r - 1$ represent indexes of sample points along range, $n_{dp} = 0, 1, \dots, N_{dp} - 1$ denote indexes of sample points along Doppler. Sequentially, we can estimate covariance matrices along range or Doppler, and then determine their eigenvalues and eigenvectors so as to extract discriminant features. Let $X = [\chi_1 \chi_2 \cdots \chi_{N_{dp}}] = [\tilde{\chi}_1 \tilde{\chi}_2 \cdots \tilde{\chi}_{N_{dp}}]^T$. Then χ_i is a sample along range, and $\tilde{\chi}_i$ is a sample along Doppler. Since a single sample $\tilde{\chi}_i$ along Doppler has enough points, we can get $\lfloor N_{dp}/2 \rfloor + 1$ new samples $\tilde{\chi}'_i$ of $\lfloor N_{dp}/2 \rfloor$ length by adding a half sample length sliding window to $\tilde{\chi}_i$, and then calculate the covariance matrix using these new samples. Because a single sample along range is much shorter, we use all of them to estimate the covariance matrix. We have

$$R_{x_r} = \frac{1}{N_{dp}-1} \sum_{i=1}^{N_{dp}} \chi_i \chi_i^H, \quad (18)$$

$$R_{x_{dp}} = \frac{1}{\lfloor N_{dp}/2 \rfloor} \sum_{i=1}^{\lfloor N_{dp}/2 \rfloor} \tilde{\chi}'_i \tilde{\chi}'_i^H. \quad (19)$$

In addition to target classification by directly using eigenvalue spectra as features, we choose four one-dimensional features to express the dispersion situations of eigenvalue spectra: let $\Lambda = [\lambda_1, \lambda_2, \dots, \lambda_{N_e}]$ be an eigenvalue spectrum, where $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_{N_e}$, and N_e is the dimension of the eigenvalue spectrum.

- The first feature: the number of larger eigenvalues that contain 98% of the eigenvalue spectrum's energy

$$f_1 = \arg_i \left(\sum_{n=1}^i \lambda_n^2 / \sum_{n=1}^{N_e} \lambda_n^2 = 0.98 \right). \quad (20)$$

- The second feature: the modified periodic factor (see [21])

$$f_2 = \varsigma \cdot \delta_{rperi}, \quad (21)$$

where δ_{rperi} is the normalized periodic factor, ς is the normalized energy entropy, and $\delta_{peri} = \lambda_1^2 / \sum_{i=1}^{N_e} \lambda_i^2$, $\delta_{rperi} = 1 - N_e \delta_{peri} / (1 - N_e)$, $\epsilon_i = E_i / \sum_{j=1}^N E_j$, $\hat{\varsigma} = - \sum_{j=1}^N \epsilon_j \log(\epsilon_j)$, $\varsigma = \hat{\varsigma} / \log(N_e)$, where E_i denotes the i th sample's energy.

- The third feature: the entropy of the eigenvalue spectrum

$$f_3 = - \sum_{i=1}^{N_e} p_i \log(p_i) = - \sum_{i=1}^{N_e} N_e \left(\lambda_i^2 / \sum_{j=1}^{N_e} \lambda_j^2 \right) \log \left(\lambda_i^2 / \sum_{j=1}^{N_e} \lambda_j^2 \right). \quad (22)$$

- The forth feature: the sum of eigenvalues normalized by the largest eigenvalue

$$f_4 = \sum_{i=1}^{N_e} \lambda_i / \lambda_1. \quad (23)$$

What we need to emphasize is that all the aforesaid analysis, formula deduction and figures are done without considering of the fuselage components' returned echoes. In reality, radar returned echoes contain fuselage component returned echoes, and what is more, the fuselage RCS is much larger than that of the rotating structures especially when they are severely shielded, leaving only the fuselage and noise echoes. In view of the fact that in the short observation time, 20–50 ms, targets can be regarded as a point traveling at a uniform velocity, the fuselage component echoes can be written as

$$S_f(t) = C_f \exp(j2\pi f_d t), \quad (24)$$

where $f_d = \frac{2v}{\lambda}$, v is the projection of the fuselage velocity onto the radar line-of-sight, i.e., $v = v_f \cos \varphi$, where v_f is the target's velocity, φ is the angle between the radar line-of-sight and v_f . The fuselage component returned echoes' amplitudes are independently distributed, and we assume that their RCS follows Swerling fluctuation model I, which means that the fuselage echo amplitude between pulses are Rayleigh distributed. Furthermore, for the fuselage echoes within pulses, we presume that their amplitudes are constant since they are strongly correlated, and the noise covariance matrix is a tridiagonal matrix because the noise is also correlated to a certain degree. Since the noise tridiagonal covariance matrix will have the same effect trend on every target's features as the unit noise covariance matrix, the eigenvalue spectra can still be used as usable features. Meanwhile, within the several tens of milliseconds for radar to observe a target, φ changes little; thus, the fuselage component returned echoes can be considered as a single frequency signal. Therefore, the radar returned echoes with fuselage component are still in line with the sum of harmonious wave model, and we can extract features by eigenvalue spectra. Figure 6 shows the four one-dimensional features' estimated probability densities of the three kinds of aeroplanes with fuselage component echoes, when radar works at PRF of 5000 Hz and with observation time of 25 ms. We can see that every feature has some discriminant capability, and in Figure 6(a), (b) and (d), the prop aircrafts are easy to be mixed with turbojets while in Figure 6(c) the turbojets are recognized well from the other aeroplanes. So there is a possibility to fuse the four features to get a better performance.

4 Performance evaluation based on data

According to eq. (1), the radar echoes consist of JEM components, fuselage component, and noise component echoes. We simulated returned echoes of the three kinds of 25 aeroplanes' under typical scene

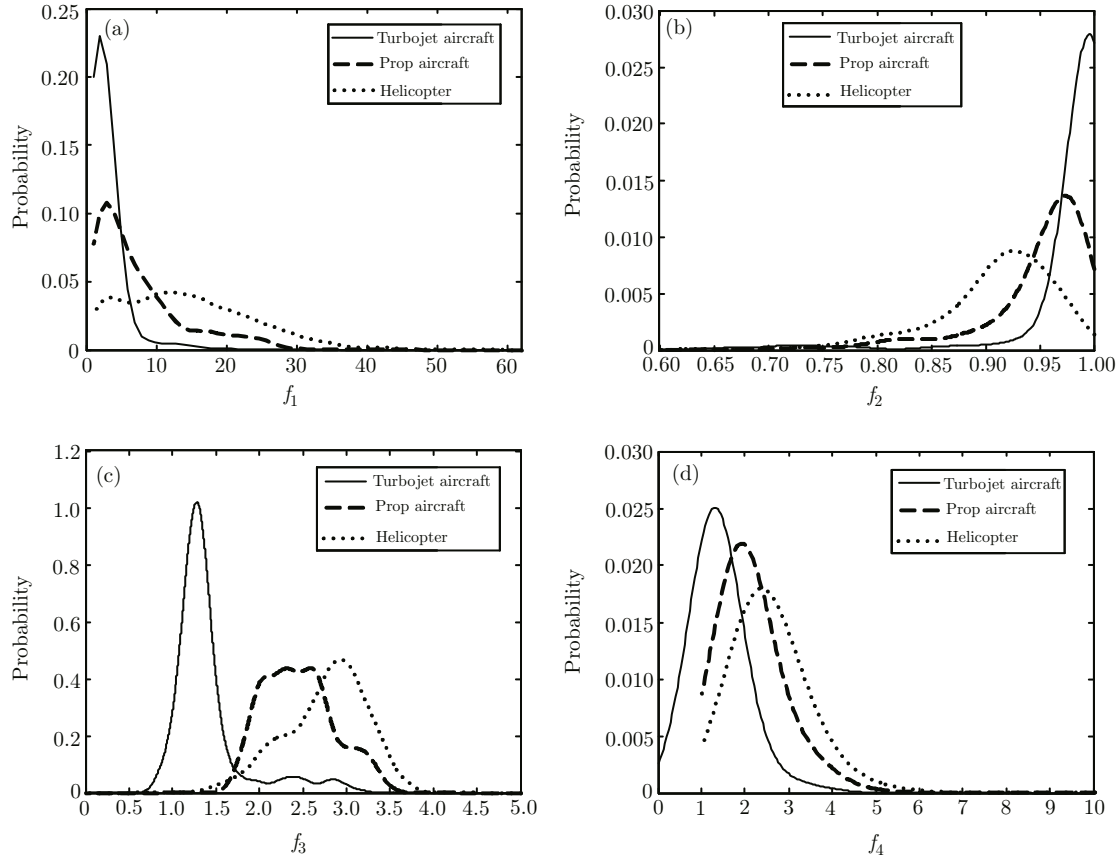


Figure 6 The four one-dimensional features' estimated probability densities of the three kinds of aeroplanes (with fuselage component echoes). (a) Number of larger eigenvalues; (b) modified periodic factor; (c) eigenvalue spectrum entropy; (d) normalized eigenvalues sum.

with eq. (2). Table 1 lists the aeroplanes' parameters, and Table 2 tabulates the scene parameters. For helicopters, because its tail rotor is much smaller in size than the main rotor, the modulated echoes are mainly affected by its own main rotor. Our simulation simply let the helicopter's tail rotor have radius (L_2) which are 1/3 of that of the main rotor, and let its corresponding rotation speed be 3 times that of the main rotor. There are random phase differences between the main and tail rotators. The blades initial angle changes within a certain range when helicopters are flying. Both the prop aircrafts and the turbojets have two rotating fans with the same configuration and rotation speed. The two fans differ in distance and random phase. Their blades angles are random within a certain range. The fuselage component echoes are $S_f(t) = C_f \exp(j2\pi f_d t)$.

Simulated classification experiment 1: We compared our feature extraction methods with two existing typical methods of using SVM (support vector machine) [22]. SVM used Gaussian kernel $K(x_i, x_j) = \exp(-\|x_j - x_i\|^2/\sigma^2)$. Since there is no a priori information about σ^2 , we just repeated the experiments for different σ^2 as many times as the computation burden could afford, and then picked the best one in this paper. The two feature extraction methods in literature are relative quantity analysis method and wave formation entropy method:

1. The relative quantity analysis method extracts a two-dimensional feature (which is corresponding to "fluctuate" in Figure 7): one-dimension is the amplitude relative quantity

$$g_a = z_n / \bar{a}, \quad (25)$$

where $\bar{a}$ is the mean amplitude of the returned echoes and z_n is the covariance of returned echoes' amplitude. The other dimension is the phase relative quantity

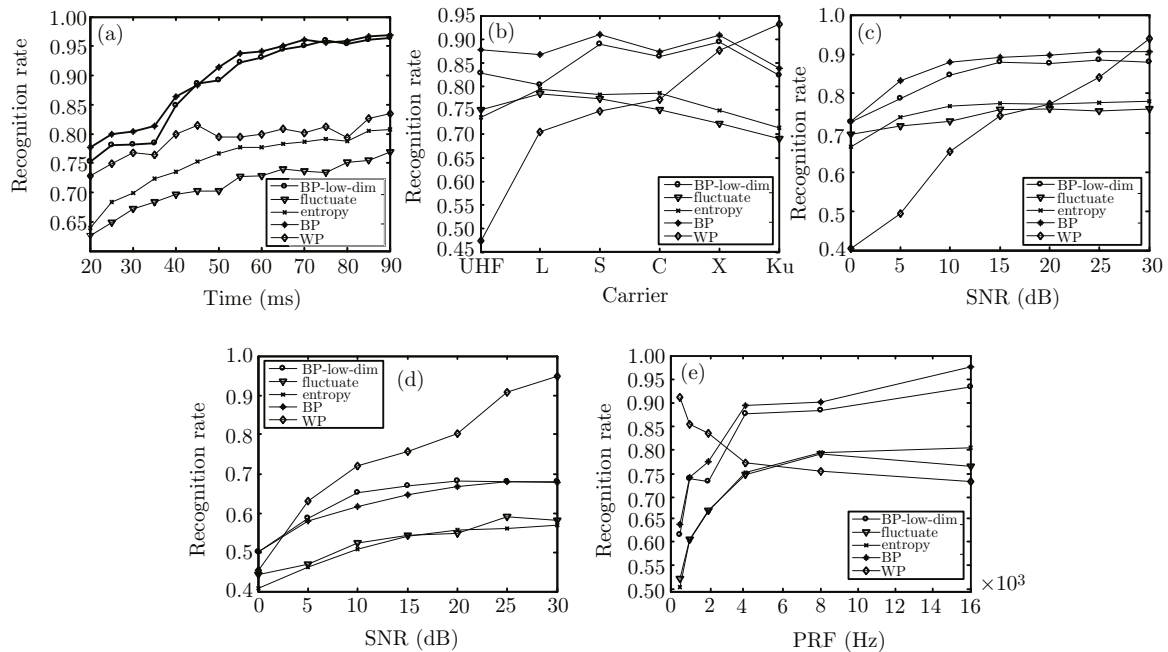
$$g_p = \arctan \left[\left(\frac{R}{R-1} \cdot \frac{1-r^2}{r^2} \right)^{0.5} \right], \quad (26)$$

Table 1 Rotation speeds and configuration parameters of the rotating structures of 6 turbojets (T), 9 prop aircrafts (P) and 10 helicopters

Aeroplane index	Rotate speed (r/min)	L_1 (m)	L_2 (m)	Blade number	Aeroplane index	Rotate speed (r/min)	L_1 (m)	L_2 (m)	Blade number
T-A	3520	0.38	1.1	38	P-H	7400	0.12	0.33	2
T-B	8615	0.18	0.51	27	P-I	7800	0.12	0.33	2
T-C	3000	0.3	1	30	H-A	132	0	16	8
T-D	5000	0.2	0.6	33	H-B	242	0	8.6	5
T-E	4000	0.24	0.8	42	H-C	192	0	10.645	5
T-F	3500	0.3	0.9	50	H-D	217	0	7.8	4
P-A	1200	0.68	1.7	4	H-E	384	0	5.5	4
P-B	1245	0.79	1.95	4	H-F	395	0	5.335	4
P-C	3400	0.27	0.76	3	H-G	406	0	5	3
P-D	3400	0.23	0.66	3	H-H	394	0	5.345	3
P-E	2340	0.23	0.66	3	H-I	324	0	7.31	2
P-F	6060	0.12	0.33	2	H-G	394	0	5.64	2
P-G	7010	0.12	0.33	2					

Table 2 Typical scenes' parameters ($U(a, b)$ is a uniform distribution on (a, b))

	Range (m)	Altitude (m)	Speed (m/s)
Helicopter	$U(30000, 40000)$	$U(500, 10000)$	$U(56, 98)$
Crop aircraft	$U(30000, 40000)$	$U(3950, 12000)$	$U(150, 200)$
Turbojet	$U(30000, 40000)$	$U(15000, 23000)$	$U(600, 700)$

**Figure 7** Single pulse series variation curves with observation time, band SNR and PRF. (a) Variation curves with time (C band, PRF of 5 kHz, SNR of 20 dB); (b) variation with carrier band (TOT of 25 ms, PRF of 5 kHz, SNR of 20 dB); (c) variation curves with SNR (C band, PRF of 5 kHz, TOT of 25 ms); (d) variation curves with SNR (L band, PRF of 500 Hz, TOT of 25 ms); (e) variation curves with PRF (C band, TOT of 25 ms, SNR of 20 dB).

where R is echo length, $r^2 = \frac{1}{(R-1)^2} [\sum_{i=1}^{R-1} \cos(\Delta\varphi_i)]^2 + \frac{1}{(R-1)^2} [\sum_{i=1}^{R-1} \sin(\Delta\varphi_i)]^2$, $\Delta\varphi_i = \varphi_i - \varphi_{i-1}$, $i = 1, 2, \dots, R-1$, φ_i is the i th echo's phase.

2. The wave formation entropy method extracts features by the following two equations (corresponding to “entropy” in Figure 7). Firstly, we normalized the wave amplitude by

$$a_1(n) = a(n) / \sum_n a(n). \quad (27)$$

Then, we obtained the wave formation entropy by

$$H = - \sum a_1(n) \log a_1(n). \quad (28)$$

Figure 7 shows the variation curves of the average recognition rates of the three kinds of aeroplanes with observation time, carrier band, the signal-to-noise rate (SNR) and PRF. (Because data used in Figure 7(a)–(e) were randomly simulated by the aforementioned model in section 3 and typical scene in section 4, and we used the same data within every figure and different data between them, it is reasonable that there exist some fluctuation between recognition rates in different figures. However, the relative relationship of the average recognition rates in all subfigures is consistent.) In Figure 7, “BP”, “BP-low-dim”, “WP” represent the eigenvalue spectra between pulses, the low dimensional eigenvalue spectra between pulse which in fact is four-dimensional and the eigenvalue spectra within pulse. It can be concluded that our features obtained better recognition performance than the other existing methods. In Figure 7(a), the average recognition rates of the five features increase as the observation time goes on, and the low-dimensional feature achieves comparable performance to the “BP”. What is more, our “BP” feature reaches average recognition rate as high as 80% when the observation time is as short as 25 ms, which is much better than the two methods available in literature. Thus, it seems that our features are more suitable for the short TOT radar. In Figure 7(b), (c), (d) and (e) we set the observation time at 25 ms. In Figure 7(b), (c) and (d), all the features except “WP” have relatively stable average recognition rates from carrier frequency to carrier frequency, and their average recognition rates increase when SNR increases, which are in accordance with the aforementioned analysis. However, Figure 7(b) also shows that the average recognition rates of eigenvalue spectra within pulse are higher for higher carrier frequency. This is due to the fact that the peaks of echoes in time domain become thinner and higher as the carrier frequency increases, which is good for the super thin sample within pulse to extract discriminative information. Nonetheless, the “WP” feature’s average recognition rates decrease quickly as the noise is raised, which will limit its application to a great extent. In Figure 7(e), under the condition of C band, observation time of 25 ms and SNR of 20 dB, when PRF is 4 kHz, the recognition rate of “BP” goes up to 90%, and after that it is almost stable. We fixed duty cycle at 3.3% to study the variation of “WP” recognition rate with PRF, and find that the higher the PRF is, the smaller the recognition rate will be, because when PRF increases the pulse number increases but the pulse width decreases. Figure 7(c) has some characteristics in common with Figure 7(d). Their observation time is 25 ms, and their recognition rates increase with increasing SNR. And for C band, 5 kHz PRF radar, “BP”’s recognition rates are higher than those of “WP”. However, for L band, 500 Hz radar, it is antithetical. Therefore, for a certain radar system, we should choose different features according to its carrier frequency, PRF, SNR, etc. Besides, we should point out that all our experiments used just one pulse series obtained during one single observation, and there still exists possibility that the recognition rates will be largely increased, if radar observe targets several times and we fuse these pulse series.

Simulated classification experiment 2: Comparison study of the classification performance of two radar systems, one of which works at C band and PRF of 5 kHz and the other works at L band and PRF of 500 Hz, when the observation time is fixed at 25 ms. Ref. [16] extracted the amplitudes of Doppler spectra using echoes between pluses after shifting the Doppler spectra to align the fuselage frequencies, then normalized the amplitudes of Doppler spectra, and reduced their dimension by PCA (principle components analysis), at last used FLD (Fisher linear discriminant) to classify targets; refs. [17, 18] had almost the same features as [16], but used LDA (linear discriminant analysis) to reduce dimensions and HMM (hidden markov model) to classify targets; ref. [19] extracted bispectra as features; ref. [23] used the fuselage Doppler frequency and the covariance around it; ref. [24] used periodicity of the time domain echoes; ref. [25] achieved JEM rotation speeds and blade number through time frequency analysis skills.

Table 3 Recognition rates of the three kinds of aeroplanes and their averages for different features (radar parameters: C band, PRF of 5 kHz, observation time of 25 ms, SNR of 20 dB)

	BP	BP-LOW-DIM	[16]	[17, 18]	[19]	[23]	[24]	[25]
Turbojet	88.02%	80.47%	53.13%	55.73%	55.99%	37.42%	26.30%	NG
Prop aircraft	84.55%	82.99%	51.39%	51.56%	65.45%	75.87%	38.37%	NG
Helicopter	73.28%	72.97%	89.53%	70.63%	85.94%	71.25%	55.94%	NG
Average recognition rate	81.95%	78.81%	64.68%	59.31%	69.13%	61.51%	40.20%	NG

Table 4 Recognition rates of the three kinds of aeroplanes and their averages for different features (radar parameters: L band, PRF of 500 Hz, observation time of 25 ms, SNR of 20 dB)

	WP	[16]	[17, 18]	[19]	[23]	[24]	[25]
Turbojet	56.25%	59.38%	50.00%	20.57%	31.51%	0%	NG
Prop aircraft	63.54%	48.78%	61.81%	64.41%	48.96%	37.67%	NG
Helicopter	91.25%	65.63%	62.19%	55.00%	47.19%	68.13%	NG
Average recognition rate	70.35%	57.93%	58.00%	46.66%	42.55%	35.27%	NG

In order to compare the performances of features in literature and our features, we chose SVM as the classifier for every kind of features so as to avoid classifier's effect on the recognition rates.

According to experiment 1, for radar working at C band and PRF of 5 kHz and with observation time of 25 ms, the "BP" feature has better recognition performance. Hence, in Table 3, we make a comparison study of eigenvalue spectra between pulses, the four-dimensional features and features in the related refs. [16–19, 23–25]. It can be concluded that our features outperform the other features. Features in [19–22] are extracted based on Doppler frequency or bispectra belonging to parametric spectra estimation methods which have defects as has been addressed before. Meanwhile, the time frequency figure obtained using the method in [25] was so bad in quality that we failed to decide the sine wave number, leading to unfinished classification. Similar to Table 3, Table 4 shows a comparison of our features within pulse with the features obtained by [16–19, 23–25], for another radar system working at L band and PRF of 500 Hz. Evidently, our features are better.

5 Conclusions

Most current low resolution radars work at low PRF and scan a target with limited time during a single sweep, which results in imprecise estimation of the characteristics when using parametric spectrum estimation methods. This paper uses the modulated JEM echoes in time domain to extract JEM features based on eigenvalue spectra. In fact, we do not accurately estimate the number of spectra contained in JEM echoes with the limitation of low PRF and short TOT; however, our features have the ability to distinguish between turbojets, prop aircrafts and helicopters from the viewpoint of pattern classification. Besides, our scheme does not require the compensation of the fuselage component. Furthermore, we propose a method to reduce computation and time burdens; that is, we use a four-dimensional vector to depict the dispersion situations of the eigenvalue spectrum though so doing sacrifices recognition rates by approximately one or two percentages. In addition, our experiments show that the features obtained between pulses or within a pulse may exchange their superiority when the radar parameters (band, PRF and SNR) change.

In this paper, we use low resolution returned echoes of narrow band radar to distinguish the three kinds of targets. Let it be noted that due to the low resolution, narrow band radar can only decide the rough sorts of targets. If we transmit several broad band signals to obtain high resolution range profiles after radar detects the target, we can probably decide the exact targets [26].

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References

- 1 Ghadaki H, Dizaji R. Target track classification for airport surveillance radar (ASR). In: Wu S J, eds. IEEE Conference on Radar 2006. Beijing: IEEE, 2006. 24–27
- 2 Zhu Z F, Ding J J. Extraction of JEM signature based decomposing the return's amplitude and phase (in Chinese). *J Air Force Radar Acad*, 2006, 20: 21–24
- 3 Leung H, Wu J. Bayesian and Dempster-Shafer target identification for radar surveillance. *IEEE Trans Aerospace Electron Syst*, 2000, 36: 432–447
- 4 Pouliguen P, Lucas L, Muller F, et al. Calculation and analysis of electromagnetic scattering by helicopter rotating blades. *IEEE Trans Antenna Propag*, 2002, 50: 1193–1408
- 5 Bell M R, Grubbs R A. JEM modeling and measurement for radar target identification. *IEEE Trans Aerospace Electron Syst*, 1993, 29: 73–87
- 6 Piazza E. Radar signals analysis and modellization presence of JEM application in the civilian ATC radars. *Aerospace Electron Syst Mag*, 1999, 14: 35–40
- 7 Martin J, Mulgrew B. Analysis of the theoretical radar return signal form aircraft propeller blades. In: *The Record of the IEEE International Conference Radar-90*. New York: IEEE, 1990. 569–572
- 8 Yang S, Yeh S. Electromagnetic backscattering from aircraft propeller blades. *IEEE Trans Magn*, 1997, 33: 1432–1435
- 9 Martint J, Mulgrew B. Analysis of the effects of blade pitch on the radar return signal from rotating aircraft blades. In: *International Conference Radar 92*. London: IET, 1992. 446–449
- 10 Elshafei M, Akhtar S, Ahmed M S. Parametric models for helicopter identification. *IEEE Trans Aerospace Electron Syst*, 2000, 36: 1242–1252
- 11 Melendez G J, Kesler S B. Spectrum estimation by neural networks and their use for target classification by radar. In: *International Conference on Acoustics, Speech, and Signal processing 1995*. New York: IEEE, 1995. 3615–3618
- 12 Moses R L, Carl J W. Autoregressive modeling of radar data with application to target identification. In: *Proceedings of the 1988 IEEE National Radar Conference*. New York: IEEE, 1988. 220–224
- 13 Pellegrini-S P F, Pardini C S. Radar signals analysis oriented to target characterization applied to civilian ATC radar. In: *International Conference Radar 1992*. London: IET, 1992. 438–445
- 14 Ding J J, Zhang X D. Studies of modulation characteristics of propeller aircraft returns in the LRR (in Chinese). *J Electron Inf Tech*, 2003, 25: 460–465
- 15 Ding J J, Zhang X D. Jet engine modulations ignatures of propeller aircraft in air-defense radar signals (in Chinese). *J Tsinghua Univ Sci Tech*, 2003, 43: 418–421
- 16 Stove A. A Doppler-based target classifier using linear discriminants and principal components. In: *The Institution of Engineering and Technology Seminar on High Resolution Imaging and Target Classification*. London: IET, 2006. 171–176
- 17 Jahangir M, Pointing K M, O'Loughlen J W. A robust doppler classification technique based on hidden markov models. In: *IEEE Radar 2002*. New York: IEEE, 2002. 162–166
- 18 Jahangir M, Ponting K M, O'Loughlen J W. Correction to Robust Doppler classification technique based on Hidden Markov models. *IEE Proceed Radar Sonar Navigat*, 2003, 150: 387
- 19 Ji H, Li J, Xie W. Bispectrum based radar target classification. In: *Fourth International Conference on Signal Processing Proceedings*. New York: IEEE, 1992. 419–422
- 20 Xu Z, Zhang K Y, Lu L. Fast algorithms of Toeplitz matrix (in Chinese). Xi'an: Northwest Polytechnic University Press, 2006. 212–255
- 21 Wang G P, Bao C C, Zhang P. Low bit rates waveform interpolation speech coding based on singular value decomposition (in Chinese). *Acta Electron Sin*, 2006, 34: 135–139
- 22 Duda R O, Hart P E, Stork D G. *Pattern Classification*. 2nd ed. New York: John Wiley and Sons, 2001. 259–264
- 23 Andric M, Durovic Z, Zrnic B. Ground surveillance radar target classification based on fuzzy logic approach. In: *International Conference on Computer as a Tool*. New York: IEEE, 2005. 1390–1392
- 24 Bullard B D, Dowdy P C. Pulse doppler signature of a rotary wing aircraft. *Aerospace Electron Syst Maga*, 1991, 36: 28–30
- 25 Yoon S, Kim B, Kim Y. Helicopter classification using time-frequency analysis. *Electron Lett*, 2000, 36: 1871–1872
- 26 Linde G. Use of wideband waveforms for target recognition with surveillance radars. In: *The record of IEEE International Radar Conference 2000*. New York: IEEE, 2000. 128–133

Appendix A [20]

Let $\mathbf{T}_n = \{\kappa_{i,j} | \kappa_{i,j} = \xi_{j-i}, \xi_{-j} = \xi_j^*\}$, $i = 1, 2, \dots, n, j = 1, 2, \dots, n$ be a Hermite Toeplitz matrix, $\mathbf{T}_k$ ($k = 1, 2, \dots, n-1$) be its k th order principle submatrix, and $P_k(\lambda) = \det(\mathbf{T}_k - \lambda \mathbf{I}_k)$, $q_1(\lambda) = \xi_0 - \lambda$, $q_k(\lambda) = P_k(\lambda)/P_{k-1}(\lambda)$, $c_k = (\xi_{-1}, \xi_{-2}, \dots, \xi_{-k})^T$, $k = 1, 2, \dots, n$.

Theorem. If λ is not one of $\mathbf{T}_{n-1}$'s eigenvalues, and $y_{n-1}(\lambda)$ is the solution vector of eq. (A1)

$$(\mathbf{T}_{n-1} - \lambda \mathbf{I}_{n-1})y_{n-1}(\lambda) = c_{n-1}. \quad (\text{A1})$$

Then

$$q_n(\lambda) = \xi_0 - \lambda - c_{n-1}^H y_{n-1}(\lambda), \quad (\text{A2})$$

$$q'_n(\lambda) = -1 - y_{n-1}^H y_{n-1}(\lambda). \quad (\text{A3})$$

And if λ is one of the eigenvalues of $\mathbf{T}_n$, its corresponding eigenvector is $x_n(\lambda) = \begin{bmatrix} -1 \\ y_{n-1}(\lambda) \end{bmatrix}$.

Proof. From $\mathbf{T}_n - \lambda \mathbf{I}_n = \begin{bmatrix} \xi_0 - \lambda & c_{n-1}^H \\ c_{n-1} & \mathbf{T}_{n-1} - \lambda \mathbf{I}_{n-1} \end{bmatrix}$, it follows that

$$\begin{aligned} p_n(\lambda) &= \det(\mathbf{T}_n - \lambda \mathbf{I}_n) \begin{bmatrix} \xi_0 - \lambda - c_{n-1}^H (\mathbf{T}_{n-1} - \lambda \mathbf{I}_{n-1})^{-1} c_{n-1} & \mathbf{0}^T \\ c_{n-1} & \mathbf{T}_{n-1} - \lambda \mathbf{I}_{n-1} \end{bmatrix} \\ &= [\xi_0 - \lambda - c_{n-1}^H y_{n-1}(\lambda)] p_{n-1}(\lambda), \end{aligned} \quad (\text{A4})$$

we get eq. (A2). And

$$(\mathbf{T}_n - \lambda \mathbf{I}_n)x_n(\lambda) = \begin{bmatrix} \xi_0 - \lambda & c_{n-1}^H \\ c_{n-1} & \mathbf{T}_{n-1} - \lambda \mathbf{I}_{n-1} \end{bmatrix} \cdot \begin{bmatrix} -1 \\ y_{n-1}(\lambda) \end{bmatrix} = \begin{bmatrix} -q_n(\lambda) \\ \mathbf{0} \end{bmatrix}. \quad (\text{A5})$$

When λ is one the eigenvalues of $\mathbf{T}_n$, $p_n(\lambda) = 0$, then $q_n(\lambda) = 0$. By (A5), $x_n(\lambda)$ is the eigenvector corresponding to λ . With the derivative of eq. (A1) with respect to λ , we get

$$(\mathbf{T}_{n-1} - \lambda \mathbf{I}_{n-1})' y_{n-1}(\lambda) + (\mathbf{T}_{n-1} - \lambda \mathbf{I}_{n-1}) y'_{n-1}(\lambda) = 0. \quad (\text{A6})$$

That is,

$$(\mathbf{T}_{n-1} - \lambda \mathbf{I}_{n-1}) y'_{n-1}(\lambda) = y_{n-1}(\lambda). \quad (\text{A7})$$

From eqs. (A7) and (A2), we know

$$q'_n(\lambda) = -1 - c_{n-1}^H y'_{n-1}(\lambda) = -1 - y_{n-1}^H(\lambda) (\mathbf{T}_{n-1} - \lambda \mathbf{I}_{n-1}) y'_{n-1}(\lambda) = -1 - y_{n-1}^H(\lambda) y_{n-1}(\lambda). \quad (\text{A8})$$